Finetune and you shall find?

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Abstract

In this project, I investigate to what extent finetuning on an information retrieval dataset could improve the search capabilities of a pre-trained Swedish sentence embedding model. This is done by machine translating the MS MARCO dataset into Swedish and running experiments inspired by previous research. Evaluation is carried out on Swedish information retrieval tasks in the MTEB benchmark. The results are non-conclusive, but show a tendency of worsening effects on one of the datasets, and a slight improvement on the other.

1 Introduction

Hybrid search that combines lexical and vector search is prevalent in modern-day information retrieval (IR) systems. The enabler for that is pre-trained embedding models, which can range in size from some million to several billion parameters. The more light-weight models have the benefit of being able to be run on CPU's, with smaller vector indices and potentially faster inference at query time.

For this project, I want to see if the general IR capabilities of a smaller Swedish sentence embedding model can be improved by fine-tuning it on some additional information retrieval data set and then evaluating it on an appropriate embedding benchmark for Swedish.

2 Related Work

Sopkovič et al. (2025) have done something similar for Slovak. They finetuned some different Sentence BERT models for Slovak by LLM-translating the MS MARCO dataset (Nguyen et al., 2016) – an IR dataset from Microsoft (discussed in more detail in section 3.2) – in order to obtain better embedding representations for Slovak information retrieval. While they do not provide any baseline, they seem to be happy with their evaluation results on a Slovak retrieval task.

3 Material

3.1 KB-SBERT

I use the `KBLab/sentence-bert-swedish-cased` model (125M parameters) from the National Library of Sweden and their KB Lab (Rekathati, 2021). Sentence transformers are a type of transformer models especially trained for semantic textual similarity (STS). Instead of following the traditional BERT approach to STS which involves passing two sentences together to the same network – a cross-encoder that produces a similarity score for them in one go – they instead use the same model with shared weights twice (a so-called siamese bi-encoder setup) to produce two different sentence embeddings, which can then be compared with cosine similarity. The important difference is that this bi-encoder approach produces embeddings which can be pre-computed and indexed for all searchable documents, and then at search time the model only needs to compute an embedding for the query.

The `KBLab/sentence-bert-swedish-cased` model, henceforth called KB-SBERT, has been trained on English-Swedish parallel corpus data using knowledge distillation. This is done by having a strong bi-encoder type teacher model produce dense embeddings for the source language sentence (in this case English). The cross-encoder type student model takes in a sentence in both the source and target language and produces two different dense embeddings, with the goal of minimizing the mean square error between these two embeddings and their teacher model's counterpart.

3.2 MS MARCO

The MS MARCO dataset from Microsoft (Nguyen et al., 2016) is actually a collection of datasets with the purpose of training better deep learning models for search. The original v1 dataset con-

tained 100 000 Bing queries with human-written answers. Sopkovič et al. (2025) use the v2 version of MS MARCO, which contains 1M such data points. Each row in the data contains a query (e.g. "why do nails get rusty"), a couple of text passages from various sources (with status 0 or 1 indicating whether they represent the desired answer), and finally an additional answer column that contains the desired text passage/answer in isolation. The answer column may also contain the value "No answer present" if no desired answer is present among the text passages. There is no Swedish translation of the MS MARCO dataset.

3.3 English-Swedish Machine Translation

Bonifacio et al. (2021) have made efforts to to translate MS MARCO into other languages using machine translation. Although Swedish is not one of them, they do provide a collection of scripts in their GitHub repository[1] for translating it into additional languages. Since their script use models provided by the Helsinki NLP research group at University of Helsinki, I investigated if the same group had any model for Swedish. As it turns out, they have made available an English-Swedish machine translation model[2], trained on parallel corpus data. I use this model for the translation of MS MARCO into Swedish.

4 Method

4.1 Pre-processing

Although Sopkovič et al. (2025) work with the much larger v2 version of MS MARCO, I decided to focus my efforts on the smaller v1 version to speed up translation and training time. However, I still follow their methodology of working with training and evaluation triplets in the following format: $\langle$*query*, *positive passage*, *negative passage*$\rangle$. In order to achieve that, I loaded the training and validation set of MS MARCO from Hugging Face and selected 1 negative passage per query and positive passage by randomly choosing one among the 9 negative/non-selected passages.

4.2 Translation

Since the translation script of Bonifacio et al. (2021) requires the input format $\langle$*id*, *sentence*$\rangle$, I cut my triplets into three different tsv files per dataset; one for the queries, one of positives and one for negatives. After applying some smaller fixes to the script's handling of the output folder and for accommodating newer versions of the dependencies (documented in commit f8da0fa of my fork[3]), I managed to translate all my files in a matter of minutes. The contents of the translated files were then re-constructed into triplets and re-loaded as training and validation dataset objects. Table 1 shows some translation examples.

English	Swedish
what age can a child go swimming alone	vilken ålder ett barn kan simma ensam
best tragedies of ancient greece	De bästa tragedierna i forntidens greece
cost to get a patent	kostnad för att få patent

Table 1: Some example of MS MARCO queries and their machine-translated Swedish counterparts.

4.3 Fine-tuning

For the finetuning itself, I use the `sentence-transformers` library from Hugging Face, taking some inspiration for the basic training arguments from a blog post by Aarsen (2024), who apply warm-up steps (in order to slowly build up to the defined learning rate) and evaluation every 100 steps. The main hyperparameters, such as learning rate (2e-5), epoch (1) and batch size (64), are taken as specified from Sopkovič et al. (2025). Sopkovič et al. (2025) also use a loss function suitable for their triplet setup, namely multiple negative ranking loss.

Multiple negative ranking ensures that for every query in the batch, the positive passage is treated as the target embedding, and the remaining positive and negative passages across the whole batch are treated as negative examples. So for a batch size of 64 (as per Sopkovič et al. (2025)), there will be 64 negative passages and 63 other positive passages treated as negative for the triplet currently being trained against. The batch sampler strategy `NO_DUPLICATES` (also taken from the blog post by Aarsen (2024)) ensures that any triplets with overlapping positive or negative passages are not handled in the same batch.

4.4 Evaluation

For evaluation, I use the MTEB benchmark (Muennighoff et al., 2022) for Scandinavian languages

[1] https://github.com/unicamp-dl/mMARCO

[2] https://huggingface.co/Helsinki-NLP/opus-mt-en-sv

[3] https://github.com/datatjej/mMARCO_fork

(Enevoldsen et al., 2025). The MTEB is a large collection of tests for measuring the performance of text embedding models on various tasks. The two tasks we are interested in here are SwednRetrieval and SweFaqRetrieval. The former is a corpus of 1,963,576 news headlines mapped to summaries of articles in Dagens Nyheter (Monsen and Jönsson, 2021). SweFaqRetrieval is a collection of almost 1000 questions and answers collected from FAQs of different Swedish authorities (Berdičevskis et al., 2023).

The main MTEB evaluation metric is normalized discounted cumulative gain at top 10 (nDCG@10). It penalizes the model if the desired target embedding is in the top 10 documents, but at a lower rank than the desired top 1 position. Target documents that fall completely outside the top 10 list results in a score of zero for that test case. The results across the whole test suite are then averaged to produce the final task result.

4.5 Baseline

Rekathati (2021) evaluate KB-SBERT on four different tasks from the SuperLim evaluation suite (Berdičevskis et al., 2023), including SweFaqRetrieval mentioned in section 4.4. The SweFaqRetrieval task is available in two different versions, and version 1 (with 513 question-answer pairs) seems to have been used by Rekathati (2021), while the MTEB benchmark seems to be using version 2 (with 976 question-answer pairs). I therefore run the KB-SBERT base model on the same task in MTEB – together with the for KB-SBERT untested task SwednRetrieval – to create a consistent baseline results, see table 2.

Task	KB-SBERT
SweFaqRetrieval	0.733
SwednRetrieval	0.706

Table 2: Baseline results for KB-SBERT on the Swedish information retrieval tasks from the MTEB benchmark.

5 Results

In table 3, the results of all experiments are summarized.

5.1 Initial experiments

I started off with finetuning KB-SBERT on the original MS MARCO dataset in English (1 epoch)

Model	Training	SweFAQ	SweDN
KB-SBERT	–	**0.733**	0.706
KB-SBERT_en	1 epoch	0.718	0.707
	3 epochs, best	0.718	0.712
KB-SBERT_sv	1 epoch	0.728	0.708
	3 epochs, best	0.725	**0.717**
	3 epochs, final	0.717	0.663

Table 3: nDCG@10 on SweFaqRetrieval (SweFaq) and SwednRetrieval (SweDN) from MTEB. *Best* is the checkpoint with the lowest validation loss; *final* is the last checkpoint (step 3738).

to sanity-check the training code and later the results on the Swedish version of the dataset. This resulted in a slight decrease for SweFaqRetrieval compared to the baseline (-0.015) and more or less no difference for the SwednRetrieval task (+0.001). I then ran the same experiment for the Swedish dataset and noticed a similar pattern; slight decrease for SweFaqRetrieval (-0.005) and more or less no change for SwednRetrieval (+0.002). But as the validation loss was still going down towards the end of the epoch (see figure 1), I decided to carry out two more experiments with an epoch size of 3.

5.2 Additional experiments

Unfortunately, the follow-up experiments with 3 epochs (all other hyper parameters the same) yielded no exceptional gains. The English run made no difference on the SweFaqRetrieval task compared to the 1 epoch training (still -0.015), although SwednRetrieval went up slightly (+0.006). When the same was done for the Swedish dataset, the best checkpoint gave a slightly bigger decrease on the SweFaqRetrieval task (-0.008) and a somewhat larger increase on the SwednRetrieval task (+0.011), producing the largest increase overall for the latter. However, upon inspecting the final checkpoint of the Swedish 3 epoch training, it was noticeable that further training with the current setup worsened the results for both tasks. The loss curves in 2 shows that the training loss goes down significantly after the first epoch, while the validation loss remains the same, suggesting overfitting.

6 Discussion

These results point in slightly different directions, raising the question to what extent the MS MARCO dataset could reasonably be expected to help with the evaluation tasks at hand. MS MARCO maps

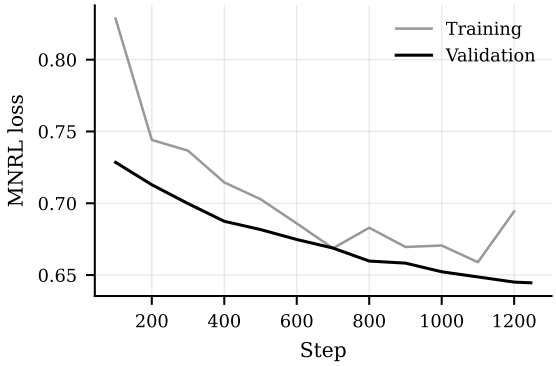


Figure 1: Training and validation loss for KB-SBERT_sv trained for 1 epoch.

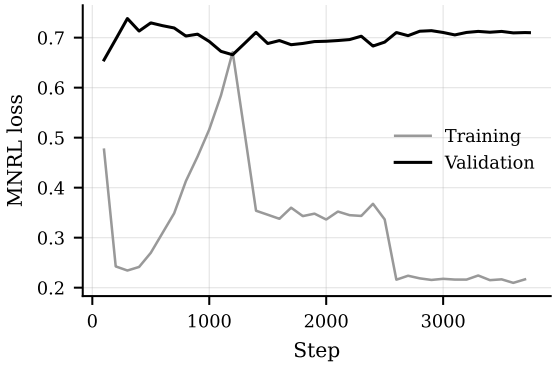


Figure 2: Training and validation loss for KB-SBERT_sv trained for 3 epochs.

queries to related search results (various domains), whereas the FAQ task maps frequently asked questions to answers (authority domain) and the DN task maps headlines to summaries (news domain). It is possible that the domains of these texts are simply a bit too different to improve the embedding weights in any decisive way.

6.1 Translation quality

A word or two should also be said about the quality of the translated dataset. The examples in table 1 show that the quality differs significantly. The first example is more or less correct although the word order is a bit awkward. In the second example, the English country name *greece* is left untranslated. The third sentence is correct. Although word order is possibly not so important for an embedding retrieval task, mistranslations could potentially decrease the quality of the finetuning. Therefore, it would have been interesting to see if a more powerful machine translation model could improve the results.

6.2 Other limitations

The timeline of the experiments did not allow for much experimentation with hyperparameter tuning. As mentioned in section 4.3, I chose the same hyperparameters as Sopkovič et al. (2025). However, they also use a significantly larger version of MS MARCO. If I were to continue these experiments, I would have tried decreasing the learning rate and added the currently unused test data split from MS MARCO as additional data.

Another limitation worth mentioning is the selection of the negative passages for each triplet. The non-selected passages of each query are still related to the same topic as the positive passage. They are not necessarily true negative as they may or may not contain the same answer to the query as the positive passage without being the preferred selection. This could introduce noise in the finetuning.

7 Conclusion

All in all, the finetuning both on English and translated Swedish data worsened the results for the SweFaqRetrieval task, while improving it slightly for the SwednRetrieval task. It is possible that the MS MARCO task is simply too different from the target evaluation tasks for finetuning to help in any significant way. However, further experiments – with possibly more data and additional hyperparameter tuning – would be needed to validate that assumption.

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